

Sida Nie

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EDUCATION

University of Wisconsin–Madison, Madison, WI [2025-present]
M.A. in Mathematics, Expected May 2027 | GPA: 4.00/4.00

Southern University of Science and Technology (SUSTech), Shenzhen, China [2022-2025]
B.S. in Mathematics, 2026 | GPA: 3.78/4.00

Joint B.S.–M.A. VISP Math Program

RESEARCH INTEREST

Riemannian Optimization, Geometric Machine Learning, Differential Geometry, Geometric Analysis

UNDERGRADUATE THESIS

[The Smooth Manifold Structure of Lie Groups](#) [Advisor : [Longting Wu](#)]

Studied the differential-geometric and topological properties of Lie groups, including left-invariant vector fields, invariant Riemannian metrics, parallelizability, Haar measure, completeness, and curvature. Explored Lie group – Lie algebra correspondence. Applied these to $SU(2)$ through explicit matrix and Lie algebra computations

SUMMER SCHOOL

Summer School on Differential Geometry [School of Mathematics Sciences, Xiamen University]

Topics: metric geometry, geometric measure theory, harmonic map

Summer School in Geometry and Analysis [Institute of Geometry and Physics, USTC]

Topics: metric geometry, general relativity

SEMINAR

Geometric Deep Learning Reading Group [UW-Madison, The RTG Mathematics & AI Synergies]

Scheduled Speaker — December 4, 2026

Reading group based on [Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges by Bronstein, Bruna, Cohen & Veličković](#)

AWARDS AND SCHOLARSHIP

SUSTech President’s Special Scholarship, Southern University of Science and Technology (2022)

Jiangxi Youth Future Academic Award (2023)

Nominated for the VISP/MA Scholarship, University of Wisconsin Madison (2026)

GRADING EXPERIENCE

The theory of Single Variable Calculus (Fernando Al Assal)

[UW-Madison]

Point set Topology (Caglar Uyanik)

[UW-Madison]

READINGS

Boumal, Nicolas. (2023). *An Introduction to Optimization on Smooth Manifolds*. Cambridge University Press.

Gallier, Jean, and Jocelyn Quaintance. (2020). *Differential Geometry and Lie Groups: A Computational Perspective*. Geometry and Computing, Vol. 12. Springer.

Zhou, Xin. (2026). *Topics in Analysis: Harmonic Map Theory*. Unpublished lecture notes, Cornell University.

Evans, Lawrence C. (1998). *Partial Differential Equations*. Graduate Studies in Mathematics, Vol. 19. American Mathematical Society.

Serre, Jean-Pierre. (1977). *Linear Representations of Finite Groups*. Graduate Texts in Mathematics, Vol. 42. Springer-Verlag. Chapter 1-5

Skills

Programming [Python, Matlab, Julia]

Typesetting [LaTeX]

Languages [English, Chinese]